

P-256 isogeny structural trait census: 1,064,000 curves

Status: **complete and independently replayed under the v1 census schema; class-wide conductor obstruction proved; ECDLP advantage unset**

Date: 2026-10-08

Result class: **exact structural census of the frozen certified population, not an exhaustive list of all mathematical properties, not all P-256 isogenies, and not an index-calculus speedup**

Outcome

The native census derived one trait record for each of the **1,064,000** previously certified, pairwise-distinct P-256-isogenous curves. Independent source replay reconstructed all **2,983,712,323** uncompressed bytes and matched the artifact byte for byte. The terminal record-chain SHA-256 is 9368fc972d7c93c7c2c425f63092464bdda504f87ea2f9a4011e8a4c48e950b1. There are no recorded discrepancies and no unresolved direction or conductor statuses.

The strongest result is class-wide rather than sample-wide. The P-256 Frobenius discriminant is -455213823400003756884736869668539463648899917731097708475249543966132856781915

and its complete proved factorization is squarefree. Because the signed discriminant is also 1 mod 4, it is a fundamental quadratic discriminant. Thus the Frobenius order already equals the maximal order:

$$Z[\pi] = \mathcal{O}_K, \quad f_\pi = 1.$$

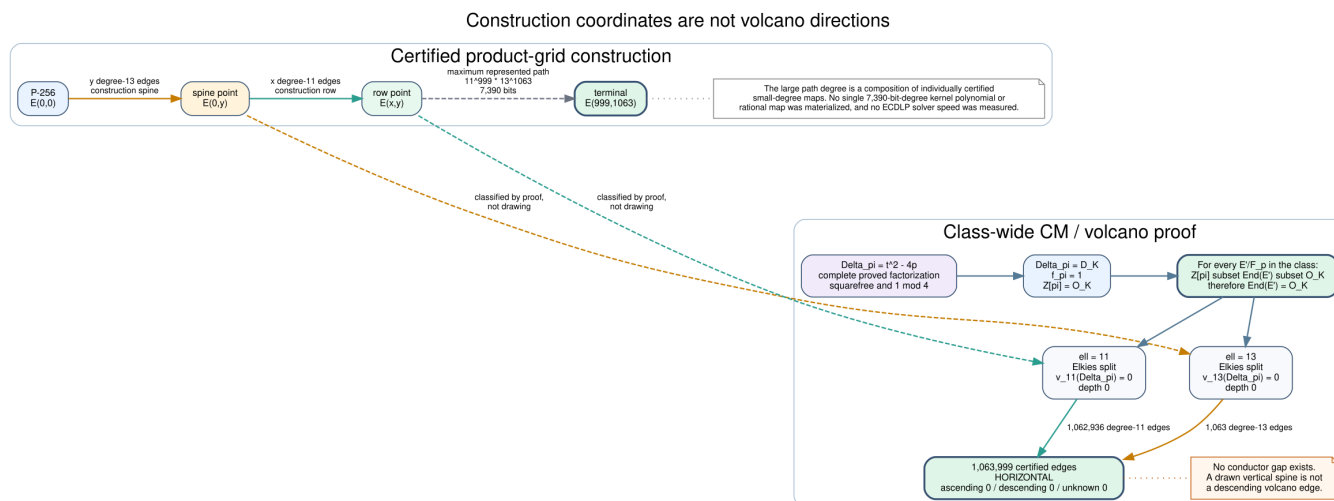
For every curve E'/F_p in the P-256 isogeny class, $Z[\pi] \subset \text{End}(E') \subset \mathcal{O}_K$; both endpoints are equal, so every such endomorphism ring is maximal. There is **no conductor gap anywhere in the F_p -isogeny class**. In particular, all **1,063,999** certified degree-11 and degree-13 edges are volcano-horizontal. Counts for ascending, descending and unknown edges are all zero.

No recorded trait establishes an easier discrete logarithm. The artifact retains `ecdlp_speedup: null`.

Requirement-to-evidence accounting

- **Census every previously certified curve — verified:** 1,064,000 inputs, unique full-width j -invariants and output records.
- **Retain class, model, path and edge traits — verified under schema v1:** deterministic JSONL plus complete byte replay.
- **Separate construction axes from volcano directions — verified:** separate fields; every certified edge is horizontal.
- **Find conductor gaps or downward routes — refuted class-wide:** the fundamental discriminant forces conductor 1.
- **Account for large factors — verified:** complete proved class-integer factorizations.
- **Find one large-prime-degree isogeny — not attempted:** source edge degrees are only 11 and 13.
- **Represent a large composite degree — verified symbolically:** $11^{999} * 13^{1063}$, 7,390 bits.
- **Exhaust 2^{32} curves or all P-256 isogenies — not achieved:** the exact population is 1,064,000.
- **Determine every possible curve property — not a finite target:** completeness is under schema v1.
- **Establish index-calculus / ECDLP speedup — not established:** no complete solver or matched ρ .

Construction axes versus volcano directions



The grid's degree-13 **spine** is drawn vertically and its degree-11 **rows** are drawn horizontally. Those are construction coordinates only. Volcano direction is determined by endomorphism orders, not by page geometry.

The largest recorded path reaches $(999, 1063)$ and has composite degree $11^{999} \cdot 13^{1063}$. Degree is multiplicative under composition, so the symbolic representation is exact. The run did not materialize one kernel polynomial or one rational map of that 7,390-bit degree, did not time point transport through the chain, and did not search for an alternative large-prime-degree route.

CM-order and volcano certificate

For P-256, the exact field prime, group order and trace are:

$p = 115792089210356248762697446949407573530086143415290314195533631308867097853951$
 $N = 115792089210356248762697446949407573529996955224135760342422259061068512044369$
 $t = 89188191154553853111372247798585809583$

The native certificate verifies $t^2 \leq 4p$, $\gcd(t, p) = 1$, and $\Delta_{\pi} = t^2 - 4p$. The group order N is proved prime, so the rational group is cyclic of prime order.

The absolute Frobenius discriminant factors as

3
 * 5
 * 456597257999
 * 1428624589419343516204097
 * 46523541035814968339936406074986559003387

All five factors are proved prime and occur to exponent one; the largest is 136 bits. Small factors use deterministic 64-bit Miller-Rabin. Larger factors use recursive Pocklington $n-1$ certificates, and exact multiplication recovers the declared integer. Ten recursive large-prime certificates are embedded in the class header.

Because Δ_{π} is fundamental, $f_{\pi} = 1$. For the two executed edge degrees:

ell	$\Delta_{\pi} \bmod \ell$	splitting	$v_{\ell}(\Delta_{\pi})$	depth	rational neighbors	certified direction
11	3	Elkies / split	0	0	2	horizontal, proved
13	10	Elkies / split	0	0	2	horizontal, proved

The order inclusion and volcano interpretation follow the ordinary finite- field classification of [Waterhouse](#)

(1969), Kohel's 1996 thesis *Endomorphism rings of elliptic curves over finite fields*, and Sutherland's [isogeny-volcano treatment](#). The last source is also available as [arXiv:1208.5370](#).

FactorDB was consulted after protocol freeze to discover candidate factors. It is not the acceptance oracle: the artifact independently proves primality and exact products in native Rust. Its query links for the two nontrivial factored values are the [Frobenius discriminant](#) and [twist order](#).

Exact population counts

Exact structural counts across 1,064,000 certified curves

Linear bars use the population of their group; zero categories remain visible as outlined markers.

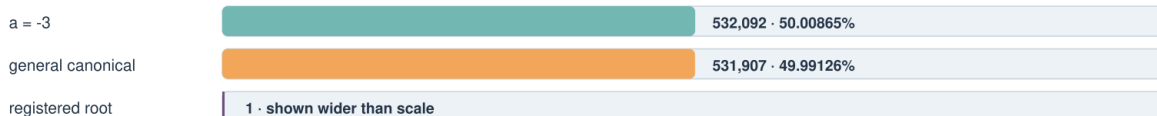
Volcano direction · 1,063,999 incoming edges



Endomorphism conductor · 1,064,000 curves



Canonical model branch · 1,064,000 curves



Special values and discrepancies



Class-wide proof: fundamental Frobenius discriminant $\Rightarrow f_{\pi} = 1 \Rightarrow$ no conductor level below the surface.

population	exact count
curves / unique j-invariants	1,064,000 / 1,064,000
incoming edges	1,063,999
degree-11 row edges	1,062,936
degree-13 spine edges	1,063
horizontal / ascending / descending / unknown	1,063,999 / 0 / 0 / 0
conductor-one / unresolved curves	1,064,000 / 0
registered-root / a=-3 / general models	1 / 532,092 / 531,907
j=0 / j=1728	0 / 0
discrepancies	0

The a=-3 and general branches occupy 50.0086466% and 49.9912594% of the population, respectively; the registered root is the remaining single record. The row count is 1,064 complete rows times 999 incoming degree-11 edges; the spine count covers global coordinates 1 through 1,063.

Field-character structure and anomaly screen

The exact quadratic-character totals are:

value	nonresidue	zero	residue
a	531,907	0	532,093
b	533,126	0	530,874
j	531,907	0	532,093
$4a^3+27b^2$	1,064,000	0	0
c4	531,907	0	532,093
c6	533,126	0	530,874

The all-nonresidue denominator initially looks anomalous, but it is forced by the group structure. The odd prime order gives no rational 2-torsion, so x^3+ax+b has no root and is irreducible. Frobenius acts on its three roots as a 3-cycle, hence the polynomial discriminant $-(4a^3+27b^2)$ is a square. Since P-256 has $p \equiv 3 \pmod{4}$, -1 is a nonresidue, forcing $4a^3+27b^2$ to be a nonresidue. This is a consistency identity, not evidence of a weak model.

The inherited 64-probe quadratic-residue screen has mean **31.996157895** and population variance **16.302705163**. Its range is 13 through 51, with zero records at the comparison threshold 52 or primary threshold 54. There are 12 records at least 49 and 14 records at most 15.

- Minimum qr_prefix_64 = 13: coordinate (133,1046), UID digest prefix 7079df18c4322662..., ICV1 suffix c036a1fb.
- Maximum qr_prefix_64 = 51: coordinate (935,866), UID digest prefix 0cab9d6f7b3526de..., ICV1 suffix 418b9695.
- Minimum signed b length, 233 bits: coordinate (293,3), UID digest prefix dee17fc049efad0c..., ICV1 suffix c58c6345.
- Minimum general-model signed a length, 236 bits: coordinate (186,142), UID digest prefix 5864c4b56e6d6a6a..., ICV1 suffix 3b7c1fdd.

The full UIDs and identities remain in the byte-replayed curve records.

These extrema are descriptive. None crosses its frozen positive-screen threshold, and no relation-generation or solver inference is attached to it.

Large factors are not large-degree maps

The twist order is

```
3 * 5 * 13 * 179
* 3317349640749355357762425066592395746459685764401801118712075735758936647
```

with a proved 241-bit largest prime factor. The target rational group order is itself a proved 256-bit prime. No embedding degree was found through the frozen bound 1,000.

These values answer the class-integer factor question but do not provide isogeny degrees. Likewise, the 136-bit prime factor of $\text{abs}(\Delta_\pi)$ is not a certified rational isogeny degree. The executed map certificates have prime degrees only 11 and 13, and every large path degree factors entirely over those two primes. Horizontal isogenies of other degrees may exist; this run did not enumerate or materialize them.

Independent replay and negative controls

Generation and verification agree on every population count, histogram, source binding, artifact digest, and summary field. Verification reopened the two original certificates and regenerated the complete artifact into a comparison writer rather than trusting its recorded derived fields.

control	expected result	observed
adjacent legacy grid plus continuation strip	pass	pass, 12/12 unique in fixture
overlapping source list	reject	rejected
changed source byte	reject	rejected

control	expected result	observed
changed direction byte	reject	rejected at byte replay
omitted required direction-reason field	reject	rejected by strict schema
unresolved direction with status and reason	accept as typed unknown	accepted in focused fixture
unresolved direction without reason	reject	rejected
known strong pseudoprimes	reject	rejected by deterministic-u64 fixture
recursive factor certificates	prove exact class integers	pass

Focused census tests: **5 passed, 0 failed**. Clippy passed with warnings denied, the release binary checked successfully, `git diff --check` passed, and Cairn scan/hook reported no findings before production.

Transfer and solver boundary

The companion [typed transfer assessment](#) separates the following obligations:

- **Small-map existence and execution:** supported by the replayed source certificates.
- **Prime-order subgroup preservation:** supported; 11 and 13 are coprime to N.
- **Conductor gap or vertical route:** refuted class-wide.
- **One executable 7,390-bit-degree map:** not materialized.
- **Large-prime-degree map:** not attempted.
- **Point transport, exceptional cases and recovery:** unknown / not measured.
- **Relation yield, matrix cost, individual-log descent and cold comparison with matched Pollard rho:** not attempted.

The weakest unresolved obligation remains destination solving. Faster known- scalar map evaluation would not by itself imply faster unknown-scalar recovery. No canonical ECDLP scoreboard is changed.

Execution receipt

The clean source revision embedded in the artifact is:

f343b90b19cae368afd4fac1db81302435bc8504

The release binary is 5,691,464 bytes with SHA-256:

7e3668a7538a33576de168b2df54cadb0c7e0a58205cf93b98e19fd6a72d18c9

phase	wall	user CPU	system CPU	peak RSS
generation	210.541 s	209.473 s	1.059 s	184,426,496 B
independent replay	167.653 s	166.936 s	0.710 s	184,356,864 B

Generation is measured from source audit through the final uncompressed record; replay is measured through the final byte comparison. The CLI computes the artifact digests after each core receipt interval. Host: Linux 6.18.44 x86-64, five exposed CPUs, 18,882,699,264 bytes RAM, Rust 1.98.0. The census path is streaming and single-process; no remote compute or S3 mutation occurred.

```
p256_isogeny_million trait-census \
--source p256-grid-1m.jsonl.gz \
--source p256-strip-y1000-h64.jsonl.gz \
--source-commit f343b90b19cae368afd4fac1db81302435bc8504 \
--output traits.jsonl.gz
```

```
p256_isogeny_million verify-trait-census \
--input traits.jsonl.gz \
--source p256-grid-1m.jsonl.gz \
--source p256-strip-y1000-h64.jsonl.gz
```

artifact	bytes
canonical gzip	468,711,780
exact decompression	2,983,712,323
generation receipt	6,466
replay receipt	6,491

Full SHA-256 digests:

- gzip: 46230554061c4ea0bf859978a5ce3c18d8f0fd0500a92d1c2d9449a2297c1c35
- decompression: 469664171037bd06d28479f920ad1213af4e2f702e63d86b27852976256353f3
- generation receipt: 0e7eb6b809a4b81a4fe681e2a09ed1588c61223d1fa18190cb262ad42e4c5155
- replay receipt: 1f538b43b83208957ff13d46cb184f4baf0b24d24ab6aedabbb26e874226bdd3

The gzip used 87.3044% of the frozen 512 MiB artifact cap. Minimum free temporary storage observed at completion remained above the 512 MiB floor.

The full artifact remains at /tmp/p256-isogeny-traits-20261008/traits.jsonl.gz for this task. That is local evidence, not durable publication. The managed environment reports no configured AWS out-bound identity, secret binding or destination bucket, so no S3 upload was attempted. Chat-provided access keys are not protocol inputs. Durable publication is blocked on an ambient worker identity and an explicit destination, not on artifact preparation.

Retained failure

The first launcher used unavailable /usr/bin/time and exited 127 before the census program started or an artifact was opened. Its empty stdout receipt and 60-byte stderr are retained. Their SHA-256 digests are, respectively:

```
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
3e737f6a9117cec194b02103b357f8668fae9adb86a8a8cd0bf25efab8948548
```

The failed launcher is not counted as a scientific run.

Conclusion and next obligations

Within the exact 1,064,000-curve population, every v1 trait is accounted for, every certified edge is horizontal, every endomorphism conductor is one, and no discrepancy or frozen detector hit appears. More strongly, the conductor and absence of vertical F_p routes are settled for the entire P-256 isogeny class by the fundamental discriminant proof.

What remains genuinely open is different: horizontal routes of other degrees, an executable single large-degree map, and any destination representation with lower end-to-end discrete-log cost. A follow-up large-prime-degree search must freeze its prime family and map-construction budget. A solver claim still requires a separate preregistered cold one-target experiment including setup, map transport, relation collection and verification, linear algebra, individual-log descent, recovery, independent answer checking, and matched Pollard rho.